



# Genetic and Phenotypic Heterogeneity of the *Nocardiopsis alba* Strains of Seawater

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## Abstract

This study deals with the genetic and phenotypic heterogeneity of the marine *Nocardiopsis alba* strains isolated during pre-monsoon, monsoon and post-monsoon seasons. The isolates were characterized for their morphological and biochemical attributes, growth media preferences, antibiotic susceptibility and extracellular enzyme secretion. *Nocardiopsis alba* strains were assessed against 12 different antibiotics, and the responses were expressed in terms of the multiple antibiotic resistance (MAR) number. The majority of the strains produced multiple extracellular enzymes: proteases, amylases and lipases. Further, the strains were characterized on the basis of 16S rRNA gene sequencing and the majority were identified as *Nocardiopsis alba* along with few strains of *Streptomyces lopnurensis*, *Nocardiopsis synnemataformans* and *Nocardiopsis dassonvillei*. Neighbor-joining (NJ) phylogenetic tree suggested variation among the genetically similar *Nocardiopsis alba* species. The study establishes significant heterogeneity with respect to genetic and phenotypic characteristics of the strains of *Nocardiopsis alba*. Phylogenetic tree and phenogram-based comparison reflect the heterogeneity in terms of different clustering patterns of the strains. Further, the whole genome sequence data available in the literature also confirm the observed heterogeneity. *Nocardiopsis alba* strains displayed a relatively regressive pattern of dependence on the environmental factors based on the canonical correspondence analysis plot. The study represents cultivation, characterization, phylogenetic analysis and enzymatic potential of the *Nocardiopsis alba* species of seawater origin.

## Introduction

Despite increasing emphasis on the non-cultivable approaches, the conventional cultivability of the microorganisms still holds good in the assessment of the overall microbial diversity in an ecosystem [1–4]. The cultivation of the microorganisms plays an important connecting link between the two approaches. The cultivation of the microorganisms is necessary to investigate the physiology and biochemical traits of the organisms and the ecosystem as a whole [5–7]. Gujarat has the longest coastline in the Indian subcontinent

proving immense possibilities to explore microbial diversity, their role in the ecosystem and biotechnological potential. The Gujarat coastline is largely unexplored and untapped with a particular reference to actinomycetes. The actinomycetes are highly diverse, gram-positive and spore-forming filamentous bacteria with high G + C content in nucleic acid [8, 9]. The cultivable diversity of the actinomycetes largely revolves around the colony and cell morphology, metabolic behavior, nutritional requirement, antibiotic susceptibility, synthesis of secondary metabolites, extracellular enzymes and phylogeny [10, 11]. In many instances, genetically identical organisms exhibit enormous variability in substrate utilization and growth behavior. In fact, these patterns define the adaptability and course of evolution of microorganisms. In the present study, the occurrence of the marine *Nocardiopsis alba* species in seawater of the Kachhighadi-Gujarat coast has been described. The study also signifies the metabolic behavior, nutritional preferences, antibiotic susceptibility and phylogenetic characteristics of the marine *Nocardiopsis alba* species. The influence of abiotic factors on the occurrence of *Nocardiopsis alba* strains has been described.

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## Materials and Methods

### Study Sites, Sampling and Determination of the Physicochemical Parameters

Shivrajpur/Kachhighadi, the study site, is located near Dwarka along the Gujarat Coast (22°21'N and 68°57'E). The water samples were collected during pre-monsoon, monsoon and post-monsoon seasons from the intertidal zone at low tide, usually in the morning hours. The details of sampling are mentioned in Table S1. The isolation procedure was carried out within 24 h of the sample collection, till then they were stored at low temperatures [12]. The sampling stations were same for the sample collections in different seasons. The monsoon season spreads from June to August receiving maximum rainfall of 180 mm. The methods for physicochemical analysis are mentioned in Table 1.

### Isolation of the Actinomycetes

Actinomycetes strains were isolated from the seawater using 10 different nutritional media, including ISP-1 to ISP-7, Nutrient agar, Starch Casein agar and Actinomycetes Isolation Agar [9, 10, 13, 14]. Further, the colonies of the actinomorphotypes based on their morphology were selected and subsequently transferred on its native media to obtain pure culture, with regular sub-culturing for the maintenance of pure cultures. The water samples were subjected to moist heat treatment and enrichment prior to the isolation. The water samples were diluted prior to the isolation as undiluted samples yielded lawn growth. The isolation was carried out with 0–15%, w/v, NaCl at pH 7 and 28 °C.

**Table 1** Physicochemical parameter analysis of samples collected in different seasons from Kachhighadi Site

| Abiotic factors                                               | Method                                                                                                                                                                             | Instrument                                    | Pre-monsoon | Monsoon | Post-monsoon |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------|---------|--------------|
| Temperature (°C)                                              | Thermometer                                                                                                                                                                        | Eutech PCD 650 portable                       | 29.2        | 27.03   | 24.8         |
| pH                                                            | Portable pH meter                                                                                                                                                                  | multi-parameter Instrument                    | 8.24        | 8.01    | 8.33         |
| Conductivity (mS)                                             | Portable conductivity meter                                                                                                                                                        |                                               | 55.67       | 52.01   | 55.05        |
| Salinity (ppt)                                                | Portable conductivity meter                                                                                                                                                        |                                               | 37.49       | 34.14   | 36.46        |
| TDS (ppt)                                                     | Portable conductivity meter                                                                                                                                                        |                                               | 27.83       | 26.05   | 27.52        |
| DO (mg/L)                                                     | Portable DO mete                                                                                                                                                                   |                                               | 5.70        | 5.60    | 5.81         |
| Turbidity (NTU)                                               | Digital Nephelo Turbidity meter                                                                                                                                                    | Eutech TN-100 turbidity meter                 | 21          | 57      | 17           |
| BOD <sub>5</sub> (mg/L)                                       | Winkler's method                                                                                                                                                                   | Titration<br>Winkler's method                 | 0.85        | 1.12    | 1.20         |
| Inorganic Phosphate (PO <sub>4</sub> <sup>3-</sup> -P) (mg/L) | Molybdenum blue method followed by spectrophotometric determination                                                                                                                | Shimadzu 1800<br>UV-Visible spectrophotometer | 0.074       | 0.042   | 0.100        |
| Nitrate-Nitrogen (NO <sub>3</sub> <sup>-</sup> -N) (mg/L)     | Cadmium Column Reduction followed by Nitrite Analysis                                                                                                                              |                                               | 0.151       | 0.125   | 0.182        |
| Nitrite-Nitrogen (NO <sub>2</sub> <sup>-</sup> -N) (mg/L)     | Reaction with an aromatic amine sulfanilamide then coupled with N (1-Nepthyl)-ethylene diamine dihydrochloride forms an azo dye which is spectrophotometrically measured at 543 nm |                                               | 0.081       | 0.084   | 0.070        |
| Ammonical Nitrogen (NH <sub>4</sub> <sup>+</sup> -N) (mg/L)   | Reaction with indophenol blue followed by sodium nitropruside and phenol in alkaline medium                                                                                        |                                               | 0.025       | 0.012   | 0.021        |
| Total Nitrogen (TN) (mg/L)                                    | Autoclaved oxidized and Cadmium Column reduction followed by nitrite analysis                                                                                                      |                                               | 0.525       | 0.415   | 0.632        |
| Reactive Silicate (SiO <sub>4</sub> <sup>-</sup> -Si) (mg/L)  | Determination based on yellow Silicomolybdic acid when acidified with a solution of molybdenum blue, absorbance measured at 810 nm                                                 |                                               | 0.390       | 0.795   | 0.212        |

## Characterization of the Actinomycetes

### Morphological and Metabolic Characterization

The actinomycetes strains were initially characterized for the colony and cell morphology. The metabolic characteristics were determined on the basis of various biochemical tests. The individual isolates were grown on 10 different media to assess their versatility and media preferences (Fig. S1) [15]. Thirteen different sugars were used to assess the sugar fermentation capabilities of the actinomycetes and changes in the color of the media were observed to follow the acid production (Fig. 4).

### Antibiotic Susceptibility and Secretion of the Extracellular Enzyme

Antibiotics of different modes of action were used to examine the susceptibility of the actinomycetes strains using the Kirby–Bauer test on Dodeca G-I-Plus (Hi-Media, India). Twelve antibiotics namely Azithromycin (AZM; 15 µg), Amikacin (AK; 30 µg), Gentamicin (GEN; 10 µg), Ciprofloxacin (CIP; 5 µg), Cefadroxil (CFR; 30 µg), Cefuroxime (CXM; 30 µg), Roxithromycin (RO; 30 µg), Ampicillin (AX; 10 µg), Cefotaxime (CTX; 30 µg), Cefoperazone (CPZ; 75 µg), Clarithromycin (CLR; 15 µg) and sparfloxacin (SPX; 5 µg) were included in the analysis. The extent of multi-antibiotic resistance of the organisms was expressed in terms of multi-antibiotic resistance (MAR) index. Higher MAR index reflects greater intensity of the multi-antibiotic resistance.

The actinomycetes were further screened for the production of extracellular hydrolytic enzymes: proteases, amylases and lipases on the gelatin, starch and tributyrin agar plates, respectively (Fig. S5a, b and c).

### 16S rRNA Gene-based Identification and Phylogenetic Analysis

16S rRNA genes were amplified by using forward and reverse primer -27F (5'AGAGTTTGATCMTGGCTCAG 3') and 1492R (5' TACGGYTACCTTGTTACGACTT 3') and sequenced on the automated DNA sequencer ABI3100 (Applied Biosystems) with Big Dye Terminator kit v.3.1 (Applied Biosystems). 16S rRNA genes of the organisms were sequenced and analyzed by the Basic Local Alignment Search Tool (BLAST) with the closest cultivable actinomycetes retrieved from the NCBI and EzTaxon server [7]. Phylogenetic trees were constructed by Molecular Evolutionary Genetic Analysis (MEGA) using Neighbor-Joining (NJ) method with a bootstrap value of 1000 (Fig. 1). The details of type strains (T) were taken from the LPSN (List of Prokaryotic names with Standing in Nomenclature) and

NCBI search. The 16SrRNA gene sequences were deposited in the NCBI database (Tables S2, S3 and S4) [10].

### Phenogram-Based Analysis of the Actinomycetes Diversity

Cluster Analysis (Phenogram) based on the biochemical and physiological attributes of the isolates was carried out by the unweighted pair-group average (UPGMA) algorithm and Jaccard similarity index [10, 16]. The UPGMA algorithms and Jaccard similarity index form a rooted tree based on the pairwise similarity (Fig. 2). The nearest similar characters grouped together forming a cluster. The primary results were converted into a binary matrix prior to the Phenogram construction. A comparative account of the phenogram versus phylogram-based diversity was discussed and interpreted [17]. A cluster of the organisms with similar phenotypic characteristics was assigned as 'phenon' [7, 18].

Pearson correlation pattern of the environmental parameters and related nutrient utilization capabilities were analyzed using PAST3.20 software (Fig. 3b). Effect of environmental factors on the abundance and diversity of the actinomycetes was adjudged by the canonical correspondence analysis (CCA) using PAST3.20 software (Fig. 3a) [19].

## Results

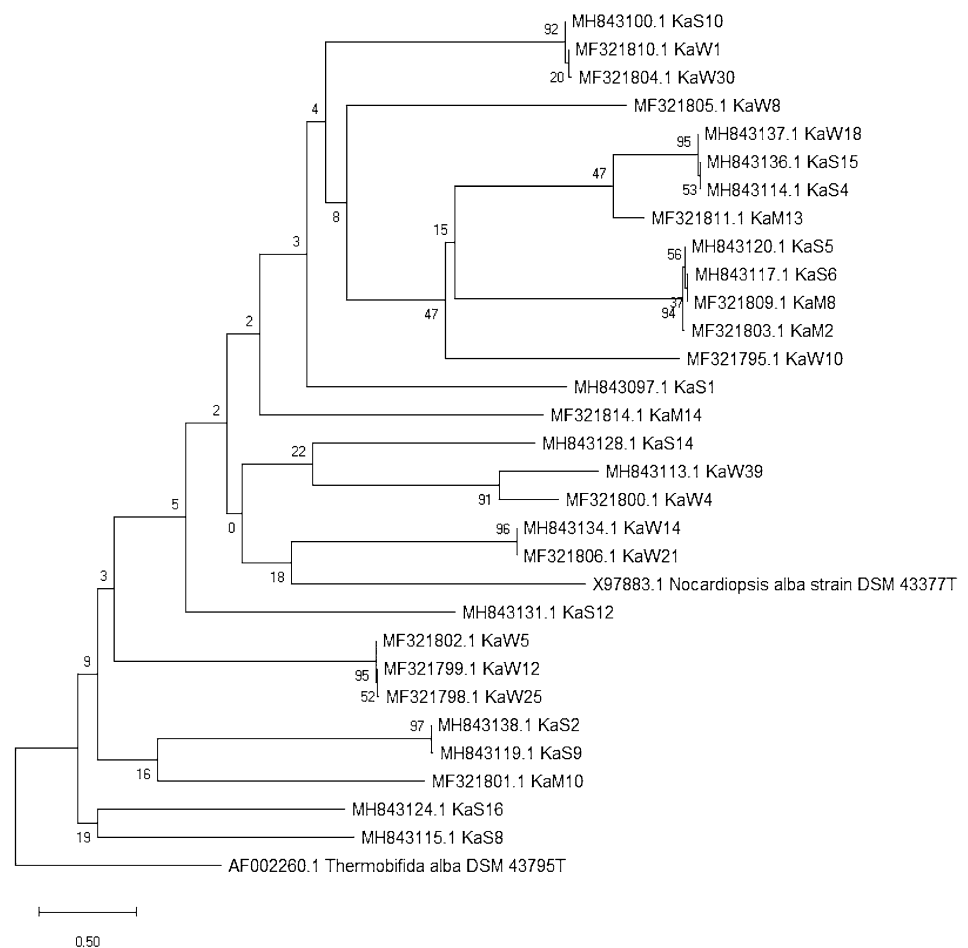
### Physicochemical Analysis of the Sample

Three seasonal samplings of pre-monsoon, monsoon and post-monsoon seasons were carried out during March 2015 to May 2016 from Kachhighadi, Gujarat Coast. Physicochemical parameters of the samples are depicted in Table 1. Significant seasonal variations in temperature, salinity, BOD, inorganic phosphate, total nitrogen content and reactive silicates were observed, while pH, electrical conductivity, total dissolved solids and dissolved oxygen reflected only marginal variation (Table 1). The temperature ranged between 24.8 °C and 29.2 °C from post-monsoon to pre-monsoon seasons. Canonical correspondence analysis represents the effect of external environmental factors on the abundance of actinomycetes strains. The effect of external environmental factors on these *Nocardiopsis alba* strains is negligible as in almost each season these strains are dominants.

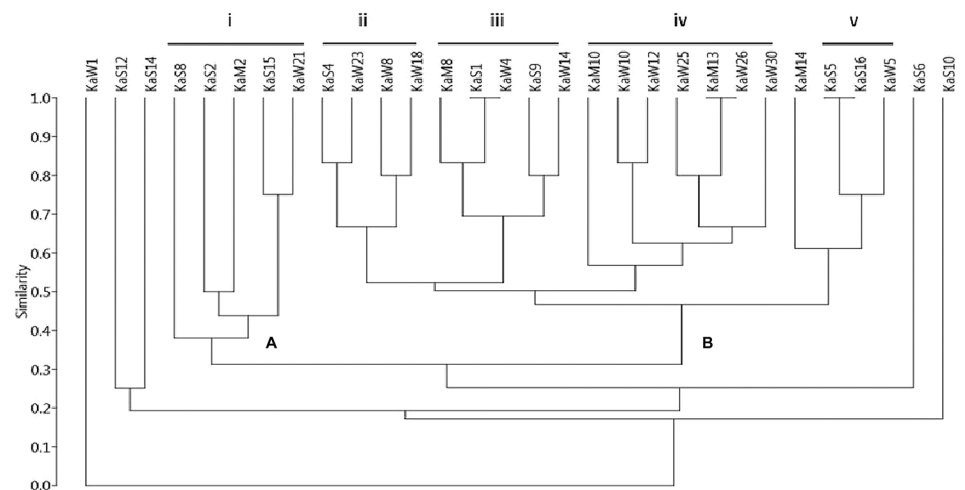
### Isolation of the Actinomycetes Using Various Approaches and Culture Media

A total of 65 actinomycetes morphotypes were obtained from pre-monsoon (16), monsoon (14) and post-monsoon (35) seasons. The untreated water samples of all three seasons yielded maximum number of the isolates while

**Fig. 1** Phylogenetic relationship among the different seasonal *Nocardiopsis alba* strains and selected types strain on the basis of 16S rRNA gene sequences. The phylogenetic rooted tree has been constructed using neighbor-joining (NJ) method. The test of phylogeny was done using bootstrap method with a value of 1000 bootstrap in MEGA software. *Thermobifida alba* DSM 43,795 species was considered as out-group for the formation of rooted tree



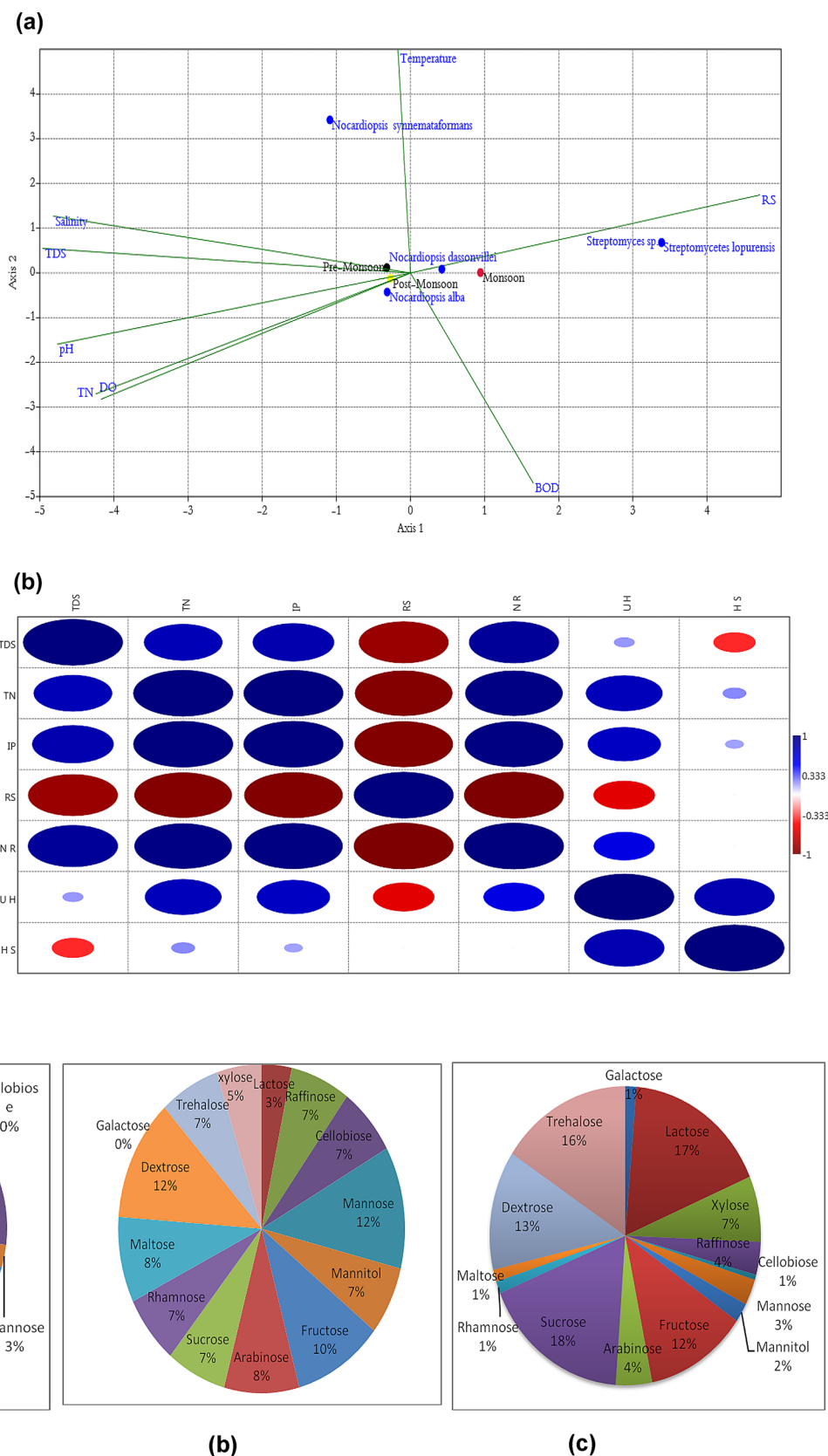
**Fig. 2** Phenotypic relationship among the different seasonal *Nocardiopsis alba* strains on the basis of morphological and biochemical characteristics. Phenogram (clusters) has been drawn using UPGMA (unweighted pair group method with arithmetic mean) method and similarity analysis was done using Jaccard similarity ( $S_j$ ) index in PAST software. The major clusters are designated as (a) and (b) while phenon is shown in roman (i, ii, iii, iv and v) numeric



heat treatment of the water samples prior to the isolation reduced the number of isolates. Enrichment of the water samples resulted in moderate counts of the actinomorphotypes [9]. A wide variation in the cultivation of the actinomycetes was evident with respect to the isolation techniques.

The majority of the isolates of the monsoon season appeared on ISP-5(Glycerol Asparagine Agar) and ISP-7 (Tyrosine agar) isolation media, followed by the Starch casein agar (SCA) medium in pre-monsoon and ISP-4 (Inorganic salt starch agar) and ISP-7(Tyrosine agar) media. Majority of the pre-monsoon isolates grew in SCA (Starch

**Fig. 3** (a) Canonical correspondence analysis (CCA) representing the effect of external environmental factors on the abundance on the actinomycetes strains. The vector (light green) represents abiotic factors while blue dotted points represent the different actinomycetes species. Black, red and yellow dots represent pre-monsoon, monsoon and post-monsoon, respectively, in the CCA plot. (b) Pearson correlation between environmental factors and their respective biochemical tests. The correlation was elucidated using PAST software. As the intensity of blue color increases, a strong positive correlation is established on other hand increased red color intensity describing strong negative correlation between the variable (Color figure online)



**Fig. 4** Sugar fermentation profile of seasonal (pre-monsoon (a), monsoon (b) and post-monsoon (c)) *Nocardiopsis alba* strains. The results are interpreted in terms of sugar utilization percentage of the strains and, respectively, indicated in the pie charts



Casein Agar), ISP-7 (Tyrosine Agar) and Actinomycetes Isolation Agar (AIA), while Nutrient Agar medium was the least preferred [9]. On the other hand, the post-monsoon morphotypes had nearly equal preferences for all the media, reflecting versatility in their metabolic behaviors.

## Morphological and Metabolic Characterization

The majority of the actinomycetes strains were chalky white and powdery in appearance with elevation and entire margin [9]. They displayed unusual characteristics on solid media like diffusible colored pigments, irregular texture and variation in substratum mycelia color (Figs. S1 and S2b). The majority of the actinomycetes in broth culture produced clumps and granulated bodies after 48hrs of growth under shaking conditions. Microscopic examinations revealed branched filaments with sporulation. The fragmentation of the filaments was observed after prolonged incubation after 72 h.

A wide range of substrate utilization was evident among the isolates of different seasons. Nearly 80% of them were capable to reduce nitrate, while litmus milk (Fig. S2c) was reduced by 75% of the isolates.

Media characterization generated a profile of the media preferences among the *Nocardiosis alba* strains (Fig. S1). The majority of the strains grew on different media after 96 h of incubation. However, a significant difference was evident with respect to the colony morphology, pigmentation, intensity of growth and size of the colony. The isolates of the pre-monsoon season preferred ISP-2 (yeast extract malt extract agar), ISP-7 (Tyrosine agar) and Nutrient agar media, while the monsoon strains preferred ISP-2, ISP-6 (peptone yeast extract iron agar) and ISP-7 media, and those of post-monsoon preferably grew on ISP-1, ISP-3 and starch agar media. A significant variation was observed among the isolates of different seasons with respect to the biochemical characteristics and sugar utilization, suggesting diverse cellular machinery. Season-wise distribution of the sugar utilization is depicted in Fig. 4.

## Antibiotic Susceptibility

The strains of different seasons have varying degrees of resistance against antibiotics (Fig. S2a). While Azithromycin (AZM; 15 µg), Amikacin (AK; 30 µg), Cefuroxime (CXM; 30 µg) and Clarithromycin (CLR; 15 µg) were the most effective antibiotics against the actinomycetes, Roxithromycin (RO; 30 µg), Ciprofloxacin (CIP; 5 µg) and Ampicillin/ Cloxacillin (AX; 30 µg) did not affect the growth. The monsoon strains; KaM 8, KaM13 and KaM 14 exhibited resistance against 8 different antibiotics, while KaM 10 was the most sensitive strain against 9 antibiotics. Further, pre-monsoon strains; KaS6, KaS9, KaS10 and KaS14 had

the highest resistance against 9 different antibiotics, while KaS1 was sensitive against 8 antibiotics. The post-monsoon strains; KaW10, KaW12 and KaW18 had resistance against 7 antibiotics, while KaW8 and KaW30 were detected sensitive against 10 antibiotics. The patterns of the antibiotics sensitivity represent diversity among the strains of different seasons.

## Extracellular Enzymes

The biotechnological relevance of the seasonal *Nocardiosis alba* strains and their ability to utilize complex substrates was adjudged by the secretion of the extracellular enzymes (Fig. S4a and b). Proteases, amylases and lipases were detected on Gelatin, Starch and Tributyrin agar plates, respectively (Fig. S5 a, b and c). Amylase secretion was predominant among the isolates while proteases were secreted by the relatively lesser number of the isolates and lipases being the least abundant. Many isolates produced a combination of these enzymes. The amylase-producing isolates were further screened on the basis of different environmental factors on starch agar plates. The optimum parameters were decided on the basis of the zone of hydrolysis on the starch agar plates. The amylase secreting isolates grew at 28–37 °C, while the optimum amylase secretion was at 28 °C. The optimum pH for the amylase secretion was 9, while the isolates did not grow at pH above 10 (Fig. S6).

## 16S rRNA Sequencing and Phylogenetic Analysis

The 16S rRNA gene sequences are presented in Tables S2, S3 and S4. Based on the 16S rRNA gene sequences, the actinobacterial isolates were identified as *Nocardiosis* and *Streptomyces*. The dominant species of all seasons were *Nocardiosis alba* followed by *Nocardiosis dassonvillei*, *Nocardiosis synnemataformans* and *Streptomyces lopnurensis*. The phylogenetic analysis of the thirty *Nocardiosis alba* strains revealed significant distinctness in terms of clustering with their type strains (Figs. 1 and S3). *Thermobifida alba* DSM 43795<sup>T</sup> was considered as out-group in the phylogenetic tree and the evolutionary relationship was established. The phylogenetic tree reveals four major groups where 13 strains out of 29 were grouped with the genetically heterogeneous groups, while the rest of the strains formed individual clusters suggesting their genetic distinctness. Isolates KaW14 and KaW21 formed a cluster with the type strain *Nocardiosis alba* DSM43377<sup>T</sup> indicating their close genetic relatedness. KaW4 and KaW39 also displayed genetic similarity with the type strain as they were grouped in the nearest cluster. The trend in the phylogenetic tree indicated that the post-monsoon strains were highly related to its corresponding type strain, while those of the pre-monsoon and monsoon seasons had greater genetic distinctness based

on the 16S rRNA homology. The strains KaW5, KaW12 and KaW25 grouped together and displayed significant phylogenetic distance from the type species. KaW10, KaM10, KaM14, KaS1, KaS14 and KaS12 formed separate branches in the phylogenetic tree indicating their distinctness from the rest of the *Nocardiopsis alba* strains of all seasons. The comparative account of the strains with the selected type strain indicated the heterogeneity among the various strains of the *Nocardiopsis alba* species.

### Analysis of the Phenogram

Phenograms represent comprehensive comparison of the microorganisms according to their phenotypic properties (Fig. 2). The genetically identical species clustered together in the phylogram displayed different clustering patterns in their corresponding phenogram indicating their phenotypic differences. Two major clusters (A and B) observed at the ~35% phenotypic similarity and five sub-clusters (i, ii, iii, iv & v) at ~50% of the phenotypic similarity are evident in the phenogram. Five isolates did not form any cluster indicating their lower phenotypic similarity. The 'Phenon' depicts clusters in the phenogram. Phenon (ii) and (v) cluster the strains of pre-monsoon and post-monsoon while phenon (iv) clusters the strains of monsoon and post-monsoon. Phenon (i) and (iii) clustered the strains of all three seasons. All the *Nocardiopsis alba* spp. were Voges–Proskauer negative. Citrate was utilized by 33% of the strains while nearly none of the strains produced indole. About 65% of the strains utilized litmus milk efficiently whereas 46% of them utilized triple sugars (glucose, lactose & sucrose) and ferrous sulfate and 63% utilized urea. The nitrate reduction was positive in 67% of the strains, while 75% produced hydrogen sulfide (Fig. S2d). Further, 27% of the strains were phenyl-alanine positive. The strains KaW1 and KaS10 formed separate branches and showed only 10 and 20% phenotypic similarity with the rest of the strains. KaS12 and KaS14 strains shared 25% similarity and clustered together in the phenogram, while KaS6 formed a different branch.

A 30% level of similarity formed two major clusters covering 25 strains indicating their phenotypic relatedness. KaS8 showed 40% of the similarity, while KaS2 and KaM2 and KaS15 and KaW21 showed 48 and 75% of the phenotypic similarity, respectively. KaM14 and KaM10 formed separate branches in their corresponding clusters with a similarity level of 65% and 56%, respectively. Further, the branches and clusters with similarity level of 62% (KaW30), 73% (KaW5), 78% (KaW25; KaW8 & KaW18; KaS9 & KaW14) and 83% (KaS4 & KaW23; KaM8; KaW10 & KaW12) are observed in the phenogram. The strains KaS1 and KaW4; KaM13 and KaW26 and KaS5 and KaS16 shared 100% similarity in the phenotypic characteristics.

### Canonical Correspondence Analysis

Analysis of the Canonical Correspondence (CCA) plots revealed that the environmental factors; salinity, pH, BOD, reactive silicates and nitrogen contents did not play any significant role in the abundance of *Nocardiopsis alba* (Fig. 3a). Further, the dominance of this species remained unaffected by the seasonal changes [20].

### Analysis of Phylogenetic and Phenotypic Heterogeneity Among *Nocardiopsis alba* spp.

Significant heterogeneity was observed between the phylogenetic and phenotypic characteristics as reflected by the phylogram and phenogram (Figs. 1 and 2). In many instances, the phylogenetic clustering did not correspond with the phenogram patterns. KaS6 grouped with two other strains in the phylogenetic tree, while it formed an individual separate branch in the phenogram. This isolate had only 28% phenotypic similarity with the rest of the isolates. A similar trend was observed with the isolates KaS10 and KaW1 which showed 10% and 20% phenotypic similarity with the rest of the strains in phenogram, while phylogenetically these isolates shared limited genetic distinctness. Furthermore, KaS1, KaW4; KaM12, KaW26 and KaS5, KaS16 shared close phenotypic similarity, while these strains are phylogenetically distinct.

### Discussion

This study analyzes the phenotypic characteristics and phylogenetic status of the *Nocardiopsis alba* strains from the seawater of Kachhighadi, Gujarat Coast. The isolates displayed variability with respect to cell morphology, filament formation and fragmentation pattern. The colonies on the solid media varied and, in some cases, violet diffusible pigments were visible when grown on specific media [21, 22]. In some cases, the cell mass appeared in the form of micro-colonies after two months of growth. The role of co-metabolism in composting and biodegradation was observed where specific microorganisms were initially able to grow with the complex materials [23, 24].

The trends of morphological variation, prolonged incubation time and halophily observed in this study are comparable with the type strain, *Nocardiopsis alba* strain DSM 43377. The reference phenotypic data of the type strains have been taken from the BacDive (the bacterial diversity metadatabase) platform (<https://bacdive.dsmz.de/>).

The preference of these organisms for a wide range of nutrients and inorganic salts indicates a blueprint of the nutrient status of the habitat and efficient cellular machinery of the marine actinomycetes [25, 26]. Colony morphology,

pigmentation and its patterns and colony texture suggest a wide variation in the adaptive mechanisms in these marine actinomycetes [5, 6]. The aggregation of the colonies reduces the surface area, a survival strategy in harsh conditions. The isolates used wide range of substrates. Nitrate reduction, litmus milk reduction, urea hydrolysis, hydrogen sulfide (H<sub>2</sub>S) production were common features among the majority of these organisms. The type strains of *Nocardio-opsis alba* utilize urea, gelatin, starch and citrate. Certain characteristics features, such as hydrogen sulfide production, which are negative in the type strain were found positive in the strains of the present study. The metabolic tests as litmus milk reduction, nitrate reduction and indole production are not reported for the type strain but the strains in the present study were positive for these characters. The optimum growth temperature of 28 °C closely matches with the type strain.

The whole genome shotgun sequencing of *Nocardio-opsis strains* suggests that the genes responsible for the morphological variation, mycelium formation and branching pattern are present in the unique core groups of the functional homologous clusters [27, 28]. The TetR family of the transcriptional regulators, part of the core groups of the functional paralogous clusters, controls the genes involved in multidrug resistance and osmotic stress. TetR family is particularly abundant in the methanogens and microbes which are exposed to the changing environment, hence involved in the adaptation strategies. The ABC transporters clusters are involved in the uptake of nutrients and other substrate utilization. LuxR and MerR family is involved in the metabolism, detoxification and oxidative stress [27, 28]. A strong positive correlation of the total nitrogen content (TN) and total dissolved solids (TDS) of the samples with the Nitrate reduction was evident.

The samples were rich in nitrogen contents and utilization of nitrates by these strains shows a strong positive correlation in Pearson's correlation chart. *Nocardio-opsis* genus is known for nitrate reduction and urea hydrolysis in the environment (Fig. 3b) [29]. Azithromycin (AZM; 15 µg), Amikacin (AK; 30 µg), Cefuroxime (CXM; 30 µg) and Clarithromycin (CLR; 15 µg) antibiotics were most effective against the majority of the strains. On the other hand, Roxithromycin (RO; 30 µg), Ciprofloxacin (CIP; 5 µg) and Ampicillin/ Cloxacillin (AX; 10 µg) did not significantly restrict the growth. The multiple antibiotic resistance (MAR) the number was significantly lower with many effective antibiotics. The majority of the isolates displayed resistance against the antibiotics indicating their abilities to survive and grow in the natural environment [30–32]. However, the mode of resistance varied from DNA replication to protein synthesis and cell wall synthesis in the actinomycetes. The existence of higher resistance against the antibiotics suggests competition among the microorganisms of the seawater.

The abilities to produce proteases and amylases suggest the availability of the complex sugars and proteinaceous substances generated from the debris of the sea organisms [33–35]. The strains produced extracellular amylases and proteases, which can play a vital role in the biodegradation of complex sugars and proteinaceous compounds in the seawater [36]. These strains thus can augment the associative biodegradation of the organic pollutant in the marine habitat. However, further studies are required for such applications [35].

Molecular identification of the actinomycetes of all seasons of the habitat reveals limited diversity. *Nocardio-opsis alba* was dominant followed by *Nocardio-opsis synnemata-formans*, *Nocardio-opsis dassonvillei*, *Streptomyces lopnurensis* and *Streptomyces sp.* Previous studies on the Gujarat coastline also suggest the dominance of *Nocardio-opsis* genus [9, 37]. The dominance of *Nocardio-opsis alba* in the seawater may probably be due to its ability to utilize a wide range of substrates and possession of varied defense strategies. The adaptation relates to the evolution rather than the development of short-term strategies [29, 38]. The ecological role of the *Nocardio-opsis* genus is widely reported with respect to host–microbe association of sponges and other invertebrates [8, 10, 39, 40]. The isolates displaying identity on the basis of 16S rRNA genes display quite different metabolic behaviors [40–43]. 16S RNA gene-based identification confirms that the majority of the isolates belong to *Nocardio-opsis alba*. However, the phylogenetic analysis of these strains with their nearest type strains shows heterogeneity as identical species clustered in different groups. The type strains cluster in a single cluster, whereas strains reported in this study formed several clusters. The clustering pattern of the strains shows that some of the strains reported in different seasons are almost identical as they clustered in the same group. On the other hand, the *Nocardio-opsis alba* strains of the same season clustered in different groups indicating the genetic heterogeneity among them. Some of the strains grouped in separate individual clades indicating significant distinctness. The trends of the heterogeneity indicate that these *Nocardio-opsis alba* strains share genetic distinctness, which may be attributed to the environmental conditions, adaptive strategies and genetic evolution. More phenons with 100% similarity indicate a lesser phenotypic distinctness, while the lesser number of such Phenons represents a greater phenotypic diversity [7, 18].

Polyphasic approach is highly useful in assessing the microbial diversity [7, 10, 18, 40, 44]. Organisms with greater genetic similarity may play significant and variable roles in the ecosystem [45]. The phenotypic heterogeneity may be due to continuously changing external environment and gene expression under particular environmental conditions. The phenotypic heterogeneity may be beneficial to the microbial population and may allow some individuals



to sustain sudden environmental changes [22, 46]. The role of the environmental factors on the microbial diversity is earlier discussed [4, 44]. *Nocardiopsis alba* in this study was dominant species as it is well adapted to the habitat [29].

The genetically similar and closely related microorganisms also display substantial diversity in terms of their cultivable behavior due to the expression of certain genes governed by the prevailing conditions of the particular niche. The expressive traits of the microorganisms decide the fate of the speciation by switching on the adaptive mechanisms [4, 44, 45]. The dominance of *Nocardiopsis alba* appears to be due to a competitive edge, efficient sinking mechanisms, associative abilities of the microorganisms, ability to utilize a wide range of nutrients and the hydrodynamics of seawater. The hydrodynamics of seawater promotes continuous disturbances, eventually affecting the distribution and abundance of the microorganisms [47, 48].

The dominance of *Nocardiopsis alba* in the seawater was apparently independent of the abiotic factors, as reflected in the CCA plot (Fig. S5) [4, 19, 49]. Further, while the role of the environmental factors may not be precisely understood in deciding the microbial dynamics, the fluctuations and gradient of such factors would certainly be playing a significant role. The evidences suggest that the highly conserved genes may not be important in judging the phenotypic heterogeneity of the microorganisms [22]. Genes involved in metabolisms usually have more chances for variation as compared to other gene families [22, 46, 50]. The phenotypic heterogeneity suggests greater adaptability of the microorganisms against the fluctuating status of the nutrients [22, 46]. The phenotypic characteristics affect the incompatibilities and biochemical conflicts among the microorganisms. The phenotypic heterogeneity further relates to the taxonomic diversity of the novel species [50].

The phylogenetic and phenotypic heterogeneity of the cultivable and non-cultivable microorganisms of the saline habitats can elucidate and establish their ecological and biotechnological relevance [13, 28, 36, 51–53].

## Conclusion

The study described in this report related to the cultivable seawater *Nocardiopsis alba* spp. of Kachhighadi, Gujarat Coast. The impact of the abiotic factors was analyzed with respect to the diversity of *Nocardiopsis alba* in this habitat. The isolates grew on a wide range of media and differentiated on the basis of cell and colony morphology, filament structure, the temporal distribution of the spore, production of diffusible pigments, secretion of hydrolytic enzymes, biochemical properties, antibiotics resistance and phylogenetic relatedness. The study revealed significant metabolic properties of *Nocardiopsis alba*. A strong correlation between

the total nitrogen content and nitrate reduction emerged. A limited genetic heterogeneity based on the 16S rRNA gene sequences was evident at the genus level, while a significant distinctness at the strain level was observed. Parallel to the genetic diversity, the strains belonging to the same species were phenotypically distinct. The study signifies the phenotypic heterogeneity of the genetically similar *Nocardiopsis alba* species of seawater origin. The findings can serve as a baseline for future investigations on the novel lineages and biotechnological exploration of marine actinomycetes.

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**Author Contributions** DSR executed the experiment, analyzed the data and wrote the manuscript; MAS assisted in actinomycetes cultivation; SDG assisted in biochemical tests; and SPS planned and designed the experiments and critically examined the data and the manuscript.

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## Compliance with Ethical Standards

**Conflict of Interest** The authors declare that they have no conflict of interest.

**Ethical Approval** This article does not contain any study with humans or animal participant.

**Informed Consent** Informed consent was obtained from all individual participants included in the study.

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